Fairness in Recidivism Prediction

Situation

Recidivism prediction refers to the task of estimating the likelihood that a convicted criminal will reoffend after being released. This type of prediction plays a critical role in the criminal justice system, influencing decisions about bail, parole, sentencing, and rehabilitation programs. In recent years, machine learning (ML) models have been increasingly deployed to assist with these predictions, promising greater objectivity and efficiency. One prominent example is the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) tool, developed by Northpointe Inc. COMPAS uses a range of input features, including criminal history, age, and responses to interview questions, to generate risk scores intended to inform judicial decisions.

Complication

However, the deployment of COMPAS and similar tools has sparked considerable controversy. In 2016, ProPublica published an investigative article titled "Machine Bias," which analyzed COMPAS scores and concluded that the tool was racially biased. According to their analysis, Black defendants were disproportionately labeled as high risk compared to white defendants, even when they did not reoffend. Conversely, white defendants were more often mislabeled as low risk. ProPublica argued that this disparity indicated systemic racial bias embedded in the algorithm.

Northpointe Inc. responded to these claims by challenging ProPublica's methodology and interpretation. They argued that COMPAS was calibrated to achieve equal predictive accuracy across racial groups and emphasized that fairness should be understood in terms of predictive parity—meaning that individuals with the same score should have the same likelihood of reoffending, regardless of race. This rebuttal highlighted the complex and often conflicting definitions of fairness in algorithmic decision-making.

Key Questions

- Based on your analyses of the data, do you find empirical support for ProPublica's allegation and for Northpointe's rebuttal?
- Given these competing perspectives, whose side are you on—ProPublica, with its emphasis on equal error rates across racial groups, or Northpointe, with its focus on predictive parity?
- More broadly, how should fairness be defined and operationalized in the context of high-stakes decisions like recidivism prediction?

Data

The dataset for this case is "compas-scores.csv".

This dataset is based on the original COMPAS Recidivism Risk Score Data that ProPublica used in their analyses. A subsample is taken where only Caucasian and African American defendants are included, and necessary data cleaning has been done to make sure the dataset is ready for analyses (see https://github.com/propublica/compas-analysis/blob/master/Compas%20Analysis.ipynb for more details on pre-processing).

Data descriptions are as follows:

two_year_recid: Indicates whether the individual was rearrested for a crime within two years of release (1 = yes, 0 = no).

score_text: A categorical label representing COMPAS's risk assessment of the individual as "Low", "Medium", or "High" risk of recidivism.

decile_score: A numerical score from 1 to 10 indicating the COMPAS-assigned risk level, with higher scores corresponding to higher assessed risk.

race: The self-identified race of the individual.

age_cat: The individual's age grouped into categories such as "Less than 25", "25 - 45", and "Greater than 45".

c_charge_degree: The severity classification of the most recent criminal charge, typically denoted as a felony (F) or misdemeanor (M).

sex: The gender of the individual, coded as "Male" or "Female".

priors_count: The total number of prior criminal offenses on the individual’s record.